

HUBERT XU

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EDUCATION

University of Toronto
Bachelor's of Computer Science
Deans List Recipient (2023)

September 2022 - June 2027

SKILLS

Languages: Python, TypeScript, Java, C, C++, SQL
Frameworks: FastAPI, ExpressJS, Spring Frameworks, Next.js
Maching Learning: CUDA, PyTorch, NumPy, OpenCV, LangChain, MCP
Cloud: Google Cloud Platform, Microsoft Azure, AWS, Cloudflare,

EXPERIENCE

Shopify Toronto, ON
Software Engineer May 2026 – August 2026

MedMe Health (YC W21) Toronto, ON
Software Engineer May 2025 – December 2025

- Architected and deployed a production robotic process automation (RPA) system enabling bidirectional data synchronization across third-party healthcare platforms to support U.S. enterprise onboarding for Fortune 500 customers, directly enabling a **7 figure annual contract pipeline**.
- Built a resilient automation pipeline combining **vision-language models** for data extraction, UI automation, and template-based visual navigation, increasing successful happy-path workflows by **40%** and decreasing costs by **70%** while gracefully handling dynamic layouts and edge-case failures.
- Implemented end-to-end observability and performance metrics across automation services, **reducing SEV response time by 300%** and enabling proactive detection of bottlenecks impacting enterprise reliability.

UofTHacks  Toronto, ON
President May 2024 – Present

- Scaled the application platform to support **3,500+ applicants** and **800+ concurrent participants**.
- Managed and mentored a team of **10+ developers**, delegating tasks and fostering a collaborative culture.
- Designed, deployed, and operated the platform using **Next.js**, **FastAPI**, and **Cloudflare**, implementing caching, request routing, and edge optimizations to meet real-world latency and reliability constraints.
- Improved end-to-end platform performance and Core Web Vitals (LCP, TTI), contributing to a **184% increase in applicants** year-over-year.

PROJECTS

npm (Open Source Contributor)  *JavaScript*

- Identified and fixed a critical vulnerability resolution bug in **metavuln-calculator** used by **npm audit**, restoring detection for prerelease package versions and improving security coverage across the npm ecosystem.

Notely  *Python, PyTorch, OpenCV, Gemini, PyTesseract*

- Built a multimodal productivity application enabling seamless integration of typed text and digital ink within a unified canvas for structured note-taking.
- Engineered semantic search over handwritten notes and diagrams by leveraging **multimodal LLMs** for indexing and **Vision Transformers** for contextual understanding of handwriting and sketches.

UserThreads *C, Linux*

- Implemented a user-level threading library for Ubuntu Linux featuring **preemptive scheduling** and priority-based thread management.
- Designed thread lifecycle management and context switching without relying on **makecontext** or **swapcontext**, demonstrating deep understanding of low-level concurrency and OS abstractions.

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)